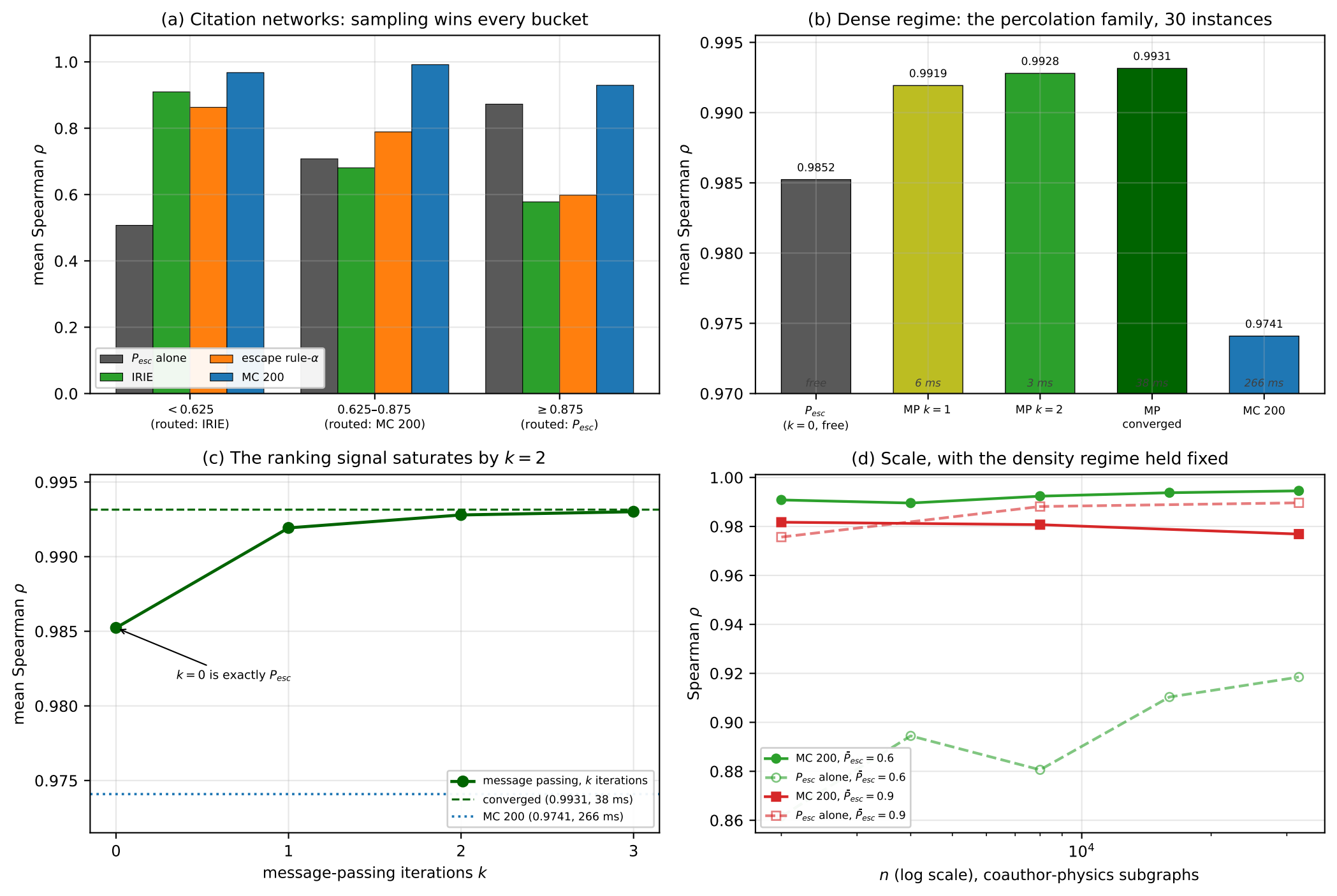
(a) Citation networks: sampling wins every bucket
mean Spearman ρ
1.0
0.8
0.6
0.4
0.2
0.0
P_{esc} alone
IRIE
escape rule-α
MC 200
< 0.625
(routed: IRIE)
0.625–0.875
(routed: MC 200)
≥ 0.875
(routed: P_{esc})
(b) Dense regime: the percolation family, 30 instances
mean Spearman ρ
0.995
0.990
0.985
0.980
0.975
0.970
0.9852
0.9919
0.9928
0.9931
0.9741
free
6 ms
3 ms
38 ms
266 ms
P_{esc}
($k = 0$, free)
MP $k = 1$
MP $k = 2$
MP
converged
MC 200
(c) The ranking signal saturates by $k = 2$
mean Spearman ρ
0.995
0.990
0.985
0.980
0.975
$k = 0$ is exactly P_{esc}
message passing, k iterations
converged (0.9931, 38 ms)
MC 200 (0.9741, 266 ms)
0
1
2
3
message-passing iterations k
(d) Scale, with the density regime held fixed
Spearman ρ
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
MC 200, $\bar{P}_{esc} = 0.6$
P_{esc} alone, $\bar{P}_{esc} = 0.6$
MC 200, $\bar{P}_{esc} = 0.9$
P_{esc} alone, $\bar{P}_{esc} = 0.9$
10^4
n (log scale), coauthor-physics subgraphs